ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Mechatronics paper 1 · practice a.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Answers must relate to the case study where asked; accept other correct answers.

SECTION A

QUESTION 1

1.1 1.1.1 D ✓ 1.1.2 A ✓ 1.1.3 C ✓ 1.1.4 C ✓ 1.1.5 A ✓ 1.1.6 D ✓ 1.1.7 B ✓ 1.1.8 D ✓ 1.1.9 B ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 False ✓ 1.2.4 True ✓ 1.2.5 True ✓ 1.2.6 True ✓ 1.2.7 True ✓ 1.2.8 False ✓ 1.2.9 True ✓ 1.2.10 False ✓ (10)

1.3 1.3.1 D ✓ 1.3.2 A ✓ 1.3.3 C ✓ 1.3.4 G ✓ 1.3.5 E ✓ 1.3.6 H ✓ 1.3.7 F ✓ 1.3.8 B ✓ 1.3.9 J ✓ 1.3.10 I ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Setpoint (desired speed) ✓ → summing junction ✓ → controller (PLC / PID) ✓ → motor drive (VSD) ✓ → motor ✓ → conveyor (process) / output speed ✓; speed sensor (encoder / tachometer) ✓ on the output with the feedback signal back to the summing junction (negative) ✓ (8)

2.2 Accuracy: open loop less accurate ✓ – closed loop accurate, corrects errors ✓; cost: open loop cheaper and simpler ✓ – closed loop more expensive (sensors, controller) ✓; disturbances: open loop cannot correct for load changes ✓ – closed loop detects and corrects disturbances automatically ✓ (6)

2.3 Correct S-shaped curve rising towards the setpoint line ✓✓; rise time marked (10%–90%) ✓; overshoot (peak above the setpoint) ✓; settling time (stays within e.g. ±2%) ✓; steady-state error (gap between final value and setpoint) ✓ (6)

[20]

QUESTION 3

3.1.1 $\mathrm{GR} = (45 \div 15) \times (60 \div 20)$ ✓ $= 3 \times 3$ ✓ $= 9:1$ ✓ (3)

3.1.2 $\mathrm{N_D} = 1\,800 \div 9$ ✓ $= 200\ \mathrm{r/min}$ ✓ (2)

3.1.3 $\mathrm{P} = 2\pi \mathrm{nT} \div 60 = 2\pi \times 1\,800 \times 5 \div 60$ ✓✓ $= 942{,}48\ \mathrm{W}$ ✓ (3)

3.1.4 $\mathrm{P_{out}} = 0{,}9 \times 942{,}48 = 848{,}23\ \mathrm{W}$ ✓; $\mathrm{T_{out}} = \mathrm{P} \times 60 \div (2\pi \mathrm{n}) = 848{,}23 \times 60 \div (2\pi \times 200)$ ✓✓ $= 40{,}5\ \mathrm{N{\cdot}m}$ ✓ (or $5 \times 9 \times 0{,}9 = 40{,}5\ \mathrm{N{\cdot}m}$) (4)

3.2 $v = \pi \times d \times n \div 60 \checkmark = \pi \times 0{,}2 \times 120 \div 60 \checkmark = 1{,}26\ m/s \checkmark$ (3)

3.3 Backlash is the free play (gap) between meshing gear teeth ✓; in robot joints it causes position errors / poor repeatability, especially when the direction changes ✓. Reduce it with anti-backlash (split) gears, harmonic drives, correct meshing / preloading ✓. (3)

[18]

QUESTION 4

4.1 At least SIX with purpose ✓✓✓✓✓✓: PLC – controls the sequence and counts; photoelectric sensor – detects a bottle arriving; stopper (pneumatic cylinder) – holds the bottle under the head; pneumatic press cylinder with 5/2 solenoid valve – presses the cap; reed switches – confirm cylinder positions; cap-presence sensor (inductive / vision) – checks the cap; conveyor motor with VSD – moves bottles; HMI / counter display; warning light / buzzer – signals a missing cap (6)

4.2 Sequence (✓ per logical step): 1. Conveyor runs. 2. Sensor detects a bottle; stopper extends and the conveyor stops (or the bottle is held). 3. Press cylinder extends to cap the bottle, then retracts (confirmed by reed switches). 4. Cap sensor checks for a cap. 5. If the cap is present, count +1, retract the stopper and release the bottle. 6. If missing, stop the station and sound the alarm; otherwise repeat. (6)

4.3 Any FOUR: compatible voltages and signal types (24 V DC I/O) ✓; mounting and alignment of sensors and cylinders ✓; cycle time / speed of each part matches ✓; safety (guards, emergency stop) ✓; hygiene requirements for food (stainless steel, washable); maintenance access; cost and available space; cable and air-line routing (4)

[16]

QUESTION 5

5.1 Any FIVE logical steps ✓✓✓✓✓: check that the air supply is on and at the correct pressure; check that the emergency stop is released and guards are closed; check the solenoid valve receives voltage (multimeter at the coil); operate the valve manually (manual override) to see if the cylinder moves; check the coil for open circuit / burnt out; check hoses for leaks or kinks; check the flow control valves are not closed; replace the faulty part, test and record the repair (5)

5.2 Breakdown: no planning cost / parts used fully ✓ – unexpected downtime, can be costly and unsafe ✓; preventive: reduces unexpected failures, planned downtime ✓ – parts may be replaced before necessary ✓; predictive: repairs only when needed, based on condition data ✓ – expensive monitoring equipment and skills needed ✓ (6)

5.3 $\text{MTBF} = \text{operating time} \div \text{number of failures} = 720 \div 4 \checkmark = 180\ h \checkmark;$
$\text{MTTR} = \text{total repair time} \div \text{number of repairs} = 12 \div 4 \checkmark = 3\ h \checkmark$ (4)

5.4 They show patterns of failures, help plan maintenance and spare parts, and prove compliance with safety laws ✓ (any one) (1)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE